NEET SAMPLE PAPER-2 2020

1. _____ is a life process that is not enential for an individuals survival but for survival of thespecies

(A) Growth (B) Reproduction (C) Respiration (D) Nutrition

2. Which of the following ART is not used in IVF program ?

(A) ZIFT (B) GIFT (C) IUT (D) All of these

3. The male and female gametophytes do not have an independent free-living existence in :

(A) Bryophytes (B) Pteridophyte (C) Gymnosperm (D) All the above

4. What is not incorrect for cockroach ?

(A) Alary muscles are 12 pairs

(B) Tubular heart with 14 chambers

(C) Chief erecretory is Uricase glands

(D) Malpighian tubules are present at the junction of foregut and mid gut.

5. Antisense technology is :-

(A) Formation of double stranded DNA from RNA

(B) A technique to prevent mRNA production from template strand of a gene of interest

(C) Introduction of RNA primer to locate a gene

(D) A technique to enhance the recombinant protein production in heterologous host

6. In multicellular organisms the mitosis occurs due to cell growth, which causes the disturbance in _____ ratio :-

(A) Nucleolus-cytoplasm

(B) Cytoplasm-organalle

(C) DNA-protein

(D) Nucleus and cytoplasm

7. Enzyme, vitamins and hormones can be classified into a single category of biological chemicals because all of these :-

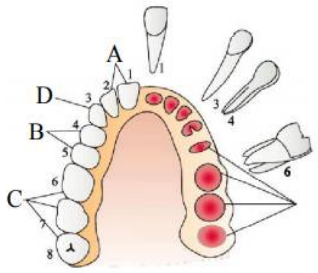
(A) enhance oxidative metabolism

(B) are conjugated proteins

(C) are exclusively synthesized in the body of a living organism

(D) help in regulating metabolism

8. In the given diagram identify A, B, C, D :-



	(A)	(B)	(C)	(D)
(A)	Incisor	Canine	Molar	Premolars
(B)	Incisor	Premolars	Molars	Canine
(C)	Premolars	Canine	Incisor	Molars
(D)	Molars	Premolars	Incisor	Canine

9. Cells forms in bone marrow include :-

(A) RBC (B) RBC and leucocytes (C) Leucocytes (D) Lymphocytes

10. Lichen and mosses do not grow in :-

(A) CO_2 polluted area (B) NO_2 polluted area

(C) CO pulluted area (D) SO_2 polluted area

11. The cells of endosperm have 24 chromosomes. What will be the number of chromosomes in gametes ?

(A) 8 (B) 16 (C) 23 (D) 32

12. All of the following represents specific epithet except :

(A) Mangifera (B) indica (C) Tuberosum (D) Leo

13. The excess of nutrients which are not used immediately are converted into _____ and stored in _____ Tissue :-

(A) Fat, Areolar

(B) Protein, Adipose

(C) Fat, Adipose

(D) Protein, white fibrous connective tissue

14. Read the following statements carefully :-

(A) Hemizygous condition can be achieved in male individuals for autosomal characters.

(B) Hybrid vigour occurs due to heterozygosity in characters.

(C) In pea plants, wrinkled seeds donot have starch grains.

(D) Chromosome 2, 3 and 6 in pea plant do not have genes for any character. How many of the above statements are wrong?

(A) A and B (B) A, B and C (C) B, C and D (D) A, C and D

15. Lysine and arginine amino acids are :

(A) (+)ively charged

(B) (–)ively charged

(C) Neural

(D) Lysine in (+)ively charged and Arginine is (–)ively charged

16. Which of the following enzyme is called regulatory enzyme of glycolysis ?

(A) Phasphogluco isomerase (B) Phaspho fructokinase

(C) Phosphotriose isomerase (D) Mutase

17. Where did australophithecinens live :-

(A) Australia

(B) East african grassland

(C) Asia

(D) Germany

18. The _____ is freely movable organ attached to the floor of the oral cavity by the frenulum.

(A) Uvula (B) Larynx (C) Tongue (D) Pharynx

19. Which of the following is an example of total stem parasite ?

(A) Rafflesia (B) Cuscuta (C) Viscum (D) Santalum

20. Which statement is not correct ?

(A) The number of cervical vertebrae are seven in almost all mammals

(B) Number of floating ribs in human are two pairs.

(C) Number of cranial bones in human are 22.

(D) Number of vertebro chondral ribs in human are 3 pairs.

21. Select the group of plants that possess stilt roots:-

(A) Zea mays, Rhizophora mangle

(B) Pandanus odoratissimus,Ficus bengalensis

(C) Rhizophora mangle, Hedera helix

(D) Ficus bengalensis, Pisum sativum

22. Which of the following represents Phylum / Division ?

(A) Primata (B) Arthropoda (C) Mammalia (D) Insecta

23. In Cockroach, each Oothecae contains eggs :-

(A) 12 – 14 (B) 14 – 16 (C) 13 – 15 (D) 16 – 18

24. Inheritance of skin colour in humans is an example of :-

(A) chromosomal aberration (B) point mutation

(C) polygenic inheritance (D) codominance

25. Resistance to jassids in cotton and cereal leaf beetle in wheat are assosiated with :

(A) Solid stem (B) High aspartic acid (C) Necter less (D) Hairy leaves

26. The first step in Dark Reaction of photosynthesis is :-

(A) Formation of ATP (B) Excitation of chlorophyll molicule

(C) Photolysis of wate (D) Attachment of carbon dioxide to pentose sugar

27. Potato and sweet potato are example of

(A) Homologous organ (B) Vestigial structure

(C) Analogous structure (D) Atavism

28. Diphyodont teeth in Human is :-

(A) 2nd premolar (B) 2nd molar (C) 1st premolar (D) 3rd molar

29. Which of the following act as 'Conduits' for energy transfer across trophic level ?

(A) Parasite (B) Predator (C) Scavanger (D) Decomposer

30. Which of the following bone is not a part of appendicular skeleton?

(A) Humerous (B) Tibia (C) Scaphoid (D) Sternum

31. In _____ placentation, a monocarpellary ovary bears a many ovules along the junction of two fused margins :-

(A) Axile (B) Parietal (C) Free central (D) Marginal

32. Only heterotrophic nutrition is found in kingdom:

(A) Monera (B) Protista (C) Fungi (D) Plantae

33. The main function of pedicellarae is :-

(A) Reproduction (B) Locomotion

(C) Cleaning (D) Food transportation

34. In a cross between individuals homozygous for (a, b) and wild type (+ +). In this cross 700 out of 1000 individuals were of parental type. Then the distance between a and b is :-

(A) 70 map unit (B) 35 map unit (C) 30 map unit (D) 15 map unit

35. Principle of plant tissue culture ?

(A) Totipotency (B) Pleuripotency (C) PCR (D) Mutation

36. Uniport, Symport and Antiport are the types of :-

(A) Simple diffusion (B) Facilitated diffusion(C) Active transport (D) Osmosis

37. Which one of the following option is least aplicable for origin of new species :-

(A) Variations (B) Genetic drift

(C) Isolation (D) Inheritane of acquired character

38. Total volume of air person can expire after a normal inspiration is :-

(A) Residual volume (B) Inspiratory capacity

(C) Expiratory capacity (D) Vital capacity

39. Species which occur most abundantly and spend their time in ecotone known as :-

(A) Key stone species (B) Endemic species

(C) Critical link species (D) Edge species

40. Regulatory proteins of skeletal muscle are :-

(A) Actin and myosin

(B) Troponin and tropomyosine

(C) Actinin and troponin

(D) Tropomyosin and myosin

41. During secondary growth in a dicot root, cork cambium is formed by the activity of :-

(A) Cortex (B) Hypodermis (C) Pericycle (D) Epidermis

42. Eubacteria cannot do :

(A) Meiosis (B) N_2 –fixation (C) Water pollution (D) Nutrient recycling

43. The common character between Balanoglossus, Herdmania and Branchiostoma is/are :-

(A) Bisexual (B) Nocturnal

(C) Retrogressive metamorphosis (D) Ciliary feeder

44. In Drosophila, how many linkage groups will be represented ?

(A) 10 (B) 8 (C) 5 (D) 4

45. Match the column I with column-II :-

Column-I	Column-II
(A) Rapeseed mustard	(i) Pusa gaurav
(B) Okra	(ii) Jassids
(C) Flat bean	(iii) Shoot borer
(D) Cow pea	(iv) Bacterial blight

(A) A-i, B-ii, C-iii, D-iv (B) A-ii, B-i, C-iv, D-iii

(C) A-i, B-iii, C-ii, D-iv (D) A-iii, B-i, C-ii, D-iv

46. Find out the correct relationship :-

(A) $\psi_S = \psi_W + \psi_P$ (B) $\psi_P = \psi_W + \psi_S$ (C) $\psi_W = \psi_P - \psi_S$ (D) $\psi_W = \psi_S + \psi_P$

47. Cancer of epithelial cells is called ?

(A) Carcinoma (B) Sarcoma (C) Leukemia (D) Lipoma

48. During swallowing entry of food into the larynx is prevented by :-

(A) Sound box (B) Trachea (C) Gullet (D) Epiglottis

49. Which productivity also known as total photosynthesis ?

(A) Gross primary productivity (B) Net primary productivity

(C) Net community productivity (D) Secondary productivity

50. A nerve impulse is transmitted from one neuron to another neuron through junction called as :

(A) Synapse (B) Neuro-muscular junction

(C) Synapsis (D) Both (A) and (B)

51. 'A' cells start division and enter in 'B' stage of meiotic division and get temporarily 'C' at this stage, called 'D' identify A, B, C and D :-

(A) A-Oogonia, B-Metaphase-I, C-Arrested, D-Primary oocyte

(B) A-Oogonia, B-Anaphase-I, C-Peleased, D-Secondary oocyte

(C) A-Oogonia, B-Prophase I, C-Arrested, D-Primary oocyte

(D) A-Oogonia, B-Telophase-I, C-Released, D-Secondary oocyte

52. All single celled eukaryotes are placed under ___(A)____, but the boundaries of this kingdom are not well defined. Identify A :

(A) Monera (B) Mycota (C) Protista (D) Plantae

53. Petromyzon differs from scoliodon in having :-

(A) Jawed mouth (B) Cranium (C) Circular mouth (D) Scales

54. The possibility of a female being haemophilic is extermely rare because mother of such a female has to be at least (a) and father should be (b)

(A) a-haemophilic, b-carrier

(B) a-carrier, b-haemophilic

(C) a-haemophilic, b-normal

(D) a-haemophilic, b-haemophilic

55. In a linear double stranded DNA 6.2 kilo base pairs are present. If thymine is 2160 then give the number of guanine base :-

(A) 8400 (B) 4040 (C) 4100 (D) 3860

56. Select the total number of incorrect statements from the following :-

(A) The diffusion rate depends on size of substrate

(B) Diffusion across membrane depends upon solubility of lipid

(C) Membrane proteins are sites for hydrophilic substance to cross membrane.

(D) Facilitated diffusion does not require concentration gradient.

(A) A (B) B (C) C (D) D

57. Incubation period of AIDS is generally ?

(A) 5–10 months (B) 10–12 years (C) 12–18 years (D) 5–10 years

58. Many collecting duct converge and through medullary Pyramids in the calyces open into the:-

(A) Renal Pelvis

(B) Duct. of Bellni

(C) Column of Bertini

(D) Vasa Recta

59. Birds and mammals attain greater body size in cold region and lesser in warm region. This rule was given by :-

(A) Allen rule

(B) Bergman rule

(C) Gause rule

(D) Dolo's rule

60. The most convulated part of hind brain is involved in :

(A) Involuntary control of voluntary muscles

(B) Voluntary control of voluntary muscles

(C) Involuntary control of involuntary muscles

(D) Voluntary control of involuntary muscles

61. Which of the following statements regarding sertoli cell is correct ?

(A) It is found in seminiferous tubule and secrete testosterone hormone

(B) It is a place where spermatozoa is concentrated and stored until ejaculation

(C) It secrete spermatozoa activating substance like fructose, citrate, inositol, prostaglandin and protein

(D) It is found in seminiterous tubule and function as nurse cells for differentiating spermatozoa

62. How many statements are correct about fungi :

(i) They shows a great diversity in morphology

(ii) They prefer to grow in warm and humid places

(iii)With the exception of yeasts which are unicellular fungi are filamentous

(iv) Only some fungi are hetetotrophs.

(A) One (B) Two (C) Three (D) Four

63. Pneumatic bones are present in :-

(A) Archaeopterix (B) Chelone (C) Corvus (D) Pteropus

64. Which of the following statement is wrong about thalassemia ?

(A) It is caused by deficient synthesis of ? and ? subunits of globin chain

(B) It is a quantitative abnormality of polypeptide globin chain synthesis

(C) Clinically the patients are anaemic and need Massive blood transfusion

(D) The patients can be treated by multidrug therapy (MDT)

65. If Meselson and stahl's experiment is continued for four generations in bacteria, The ratio of $N_{15}/N_{15} : N_{15}/N_{14} : N_{14}/N_{14}$ containing DNA in the fourth generation would be :-

(A) 1 : 1 : 0 (B) 1 : 4 : 0 (C) 0 : 1 : 3 (D) 0 : 1 : 7

66. Which of the following translocated through phloem :-

(A) Sugars (B) Amino acid (C) Hormone (D) All of these

67. Antiserum has :-

(A) Antigen (B) Antibody (C) W.B.C. (D) R.B.C.

68. Shapes of Henle's loop and vasa recta are :-

(A) 'C' shaped and 'U' shaped respectively

(B) 'U' shaped and 'C' shaped respectively

(C) Hairpin shaped and 'U' shaped respectively

(D) Hairpin shaped and 'C' shaped respectively

69. Which of the following is not an example of exhaustible resource ?

(A) Natural gas (B) Coal (C) Tidal power (D) Soil

70. Which of the following nerve helps in maintaining the equilibrium of body :

(A) Auditory nerve

(B) Abducens nerve

(C) Trochlear nerve

(D) Cochlear nerve

71. HCG is secreted by :-

(A) Inner cell mass (B) Trophoblast layer (C) Liquour folliculie (D) Chorion

72. Which of the following is known as : Contagium vivum fluidum (infectious living fluid)

(A) Eubacteria (B) Fungi (C) Viruses (D) Mycoplasm

73. Secretory unit of sebaceous gland is composed of :-

(A) Cuboidal epithelium

(B) Stratified squamous epithelium

(C) Squamous epithelium

(D) Columnar epithelium

74. Plasmid is used as vector because :-

(A) It is circular DNA which have capability to join with eukaryotic DNA

(B) It can move between prokaryotes and eukaryotes

(C) It has antibiotic resistant gene

(D) It is main part of bacterial chromosome

75. s-phase marks the phase of :-

(A) Normal size maintaining stage

(B) Organelle duplication

(C) Chromosome duplication

(D) Cytoplasmic growth

76. Which of the following expression describe nitrogen fixation :-

(A) $N_2 + 3H_2 \rightarrow 2NH_3$ (B) $2NH_4^+ + 2O_2 + 8e^- \rightarrow N_2 + 4H_2O$

(C) $2NH_3 \rightarrow N_2 + 3H_2$ (D) $2N_2$ + glucose $\rightarrow$ 2 amino acid

77. Immunity aquired after an infection is ?

(A) Active immunity (B) Passive immunity

(C) Innate immunity (D) both 2 and 3

78. Uriniferous tubules are mainly concerned with :-

(A) Concentration of urine

(B) Passage of urine

(C) Removal of urea from bloo

(D) Reabsorption of useful substances from glomerular filtrate

79. Wild life protection act in India introduced in :-

(A) 1972 (B) 1974 (C) 1988 (D) 1986

80. Which gland stores hormone in intercellular space before it's secretion into blood :

(A) Pituitary gland (B) Thyroid gland (C) Hypothalamus (D) Ovary

81. Study the given statements and answer the questions :- During 'P' phase of the menstrual cycle if pregnancy does not happen, the 'Q' dies usually around day 22 in a 28 days cycle. The drop in 'R' levels causes the lining of the uterus to fall away. This is known as 'S'. Identify P, Q, R, S:-

(A) P-menstrual, Q-Graffian follicle, R-Estrogen, S-Menstruation

(B) P-Ovulatory, Q-Endometrium, R-FSH, S-Menopause

(C) P-Luteal, Q-Corpus luteum, R-Progesterone, S-Menstruation

(D) P-Follicular, Q-Secondary oocyte, R-LH, S-Menstruation

82. Among the following how many are the mosses ? Funaria, Polytrichum, Sphagnum, Marchantia, Riccia, Equisetum, Selaginella

(A) Two (B) Three (C) Six (D) Five

83. Which one of the following option is incorrect about cockroach ?

(A) One pair of anal style are present 9th sternum of male cockroach only.

(B) One pair of anal cerci are present 10th tergum of male and female cockroach.

(C) 4 pair of malpighian tubules are found in cockroach between midgut and hind gut.

(D) Uricose glands are excretory structures found only in male cockroach.

84. Which of the following is incorrectly matched:-

(A) Gel electrophoresis – separation of charged biomolecules

(B) Competent host cell – cell ready to uptake alien DNA

(C) Biolistic method – vector mediated gene transfer

(D) DNA ligase – Molecular glue

85. After the dissolution of the synaptonemal complex, the bivalents are attached to each other only at the site of cross-overs the structure formed is add :-

(A) Synaptonemal complex (B) Synopsis

(C) Chiasmata (D) Terminalisation

86. In photosynthesis, Hydrogen is transferred from the light reaction to Dark Reaction by :-

(A) DPN (B) FAD (C) ATP (D) NADP

87. Inbreeding refers to the mating between:

(A) The animals of same breed having atleast one common ancestor upto 4-6 generations

(B) The animals of same breed having no common ancestors upto 4-6 generations on either sides of pedigree

(C) Animals of two different breed of same species

(D) Animals of two different species of same genus

88. Donor X and recipient Y belong to same blood group. Transfusion has led to RBC agglutination because :-

(A) X is Rh^-, Y is Rh^+ (B) X is Rh^+ , Y is Rh^-

(C) Both are Rh^- (D) Both are Rh^+

89. Which of the following is not an example of nonbiodegradable pollutant ?

(A) Glass (B) DDT (C) Wood (D) Compounds of phenol

90. Match the Column-I with Column-II correctly :-

I	II
(A) Oxytocin	(P) Stimulate ovulation
(B) Prolactin	(Q) Implantation and maintenance of pregnancy
(C) Luteinising hormone	(R) Lactation after child birth
(D) Progesteron	(S) Uterine contraction during labour pain

(A) A-P, B-Q, C-R, D-S (B) A-P, B-Q, C-S, D-R

(C) A-S, B-R, C-P, D-Q (D) A-S, B-R, C-Q, D-P